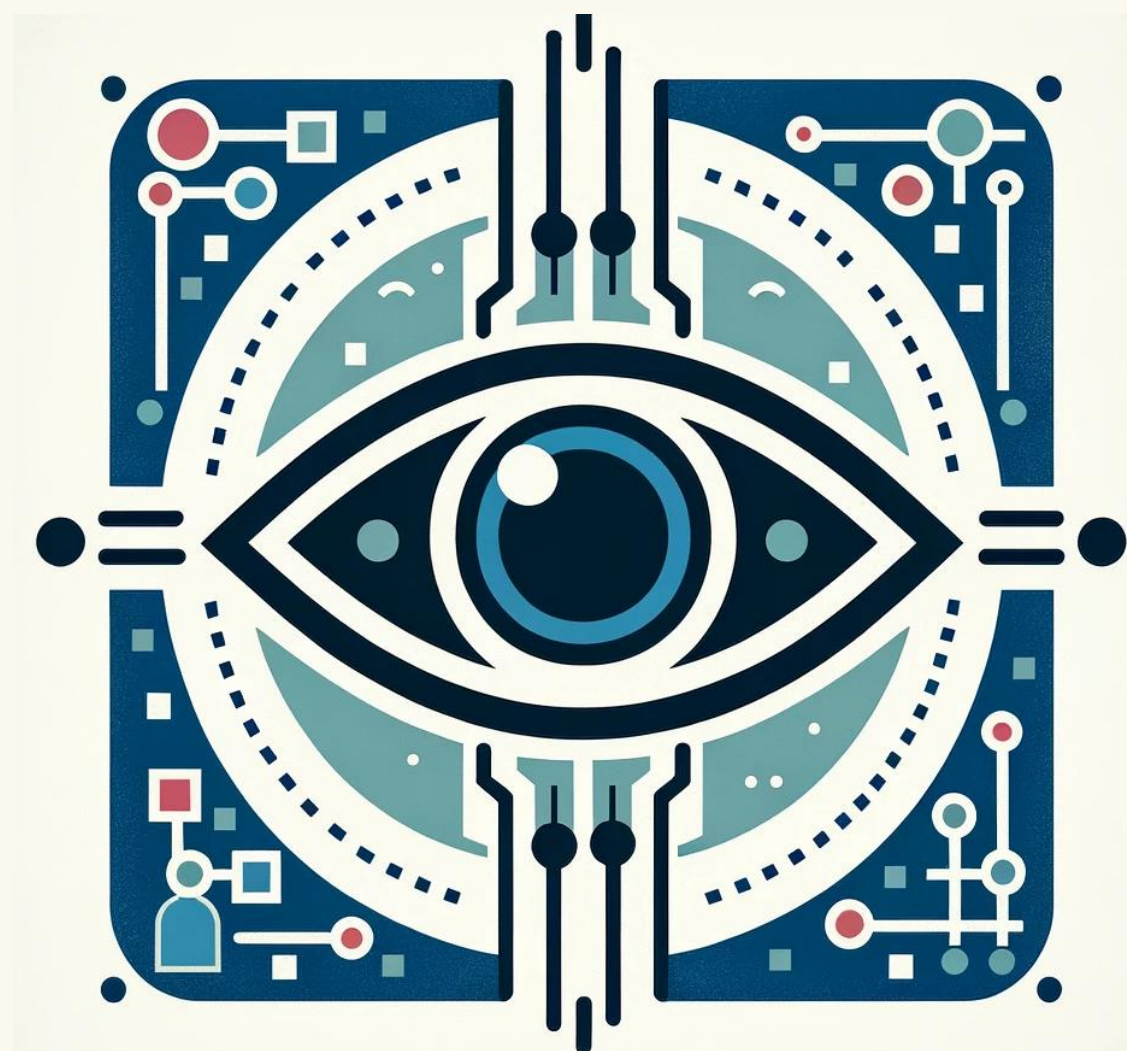




COMPUTER VISION

Recognition

Recognition

Before diving into deep learning, let us discuss how we can achieve some computer vision recognition tasks with what we know

- **Object recognition** – does a (known) object exist in the image?
 - local features extraction → matching with database
- **Scene classification** – what type of image is this?
 - image descriptor extraction → classification



Classification in computer vision

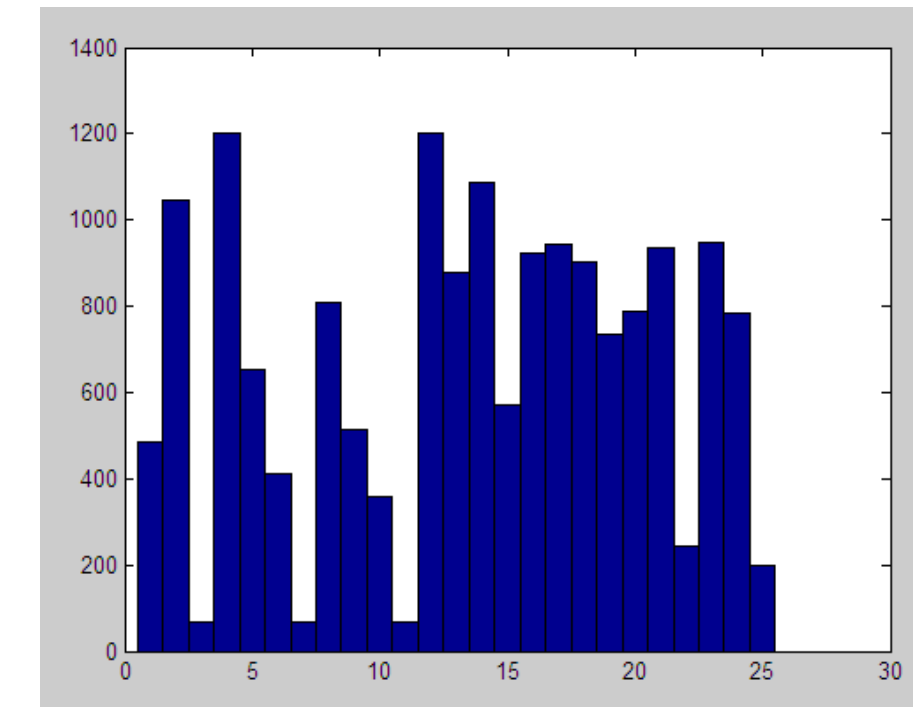
So, what do we need for image classification?

Image features

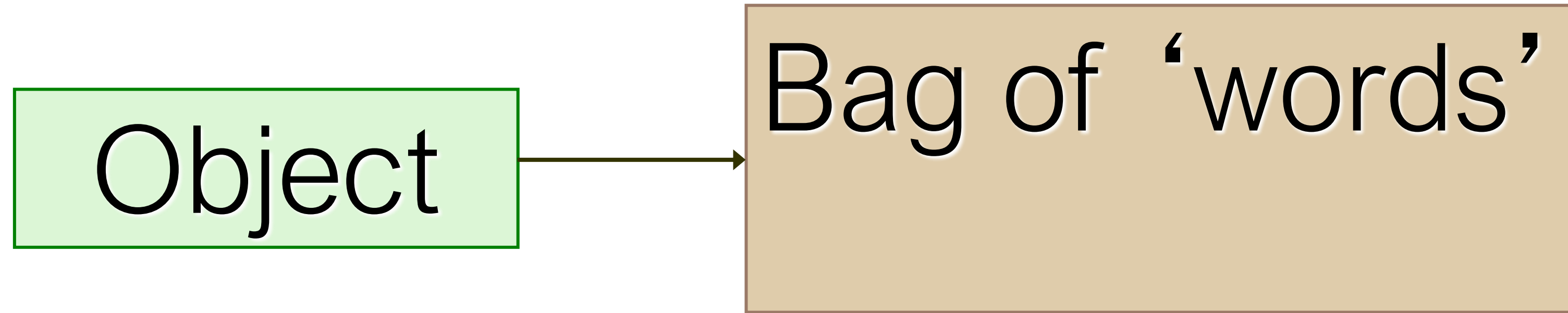
Decision model

Image features

- Raw pixels
- Hand-crafted features
 - e.g. colour, texture, shape
- Histograms
- Local descriptors
 - e.g. SIFT, ORB, HOG
- Bags of descriptors
- Local + global

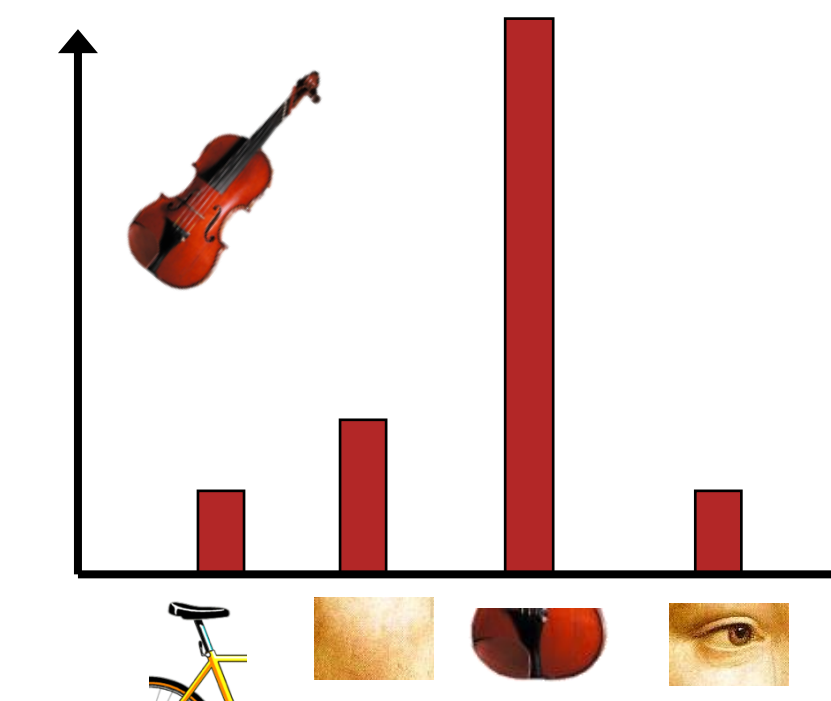
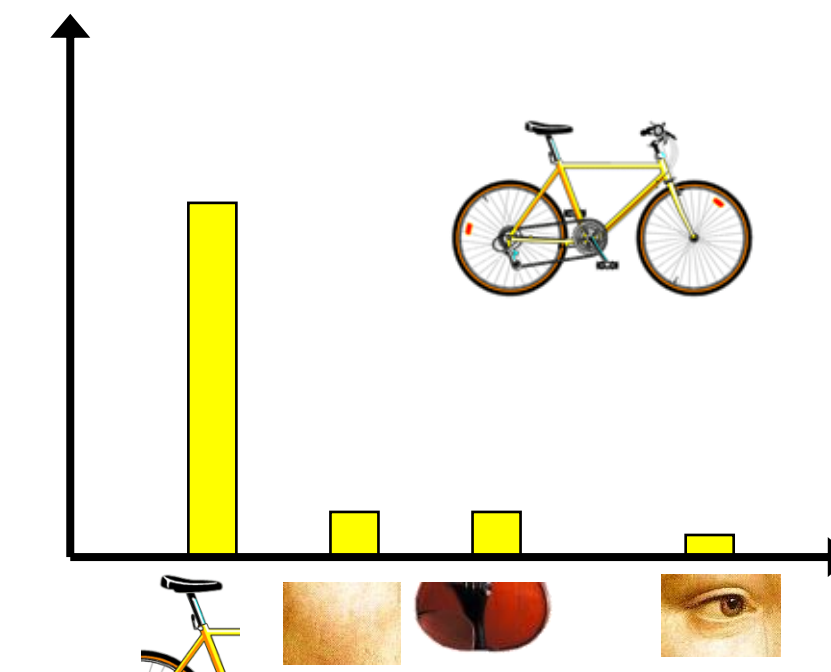
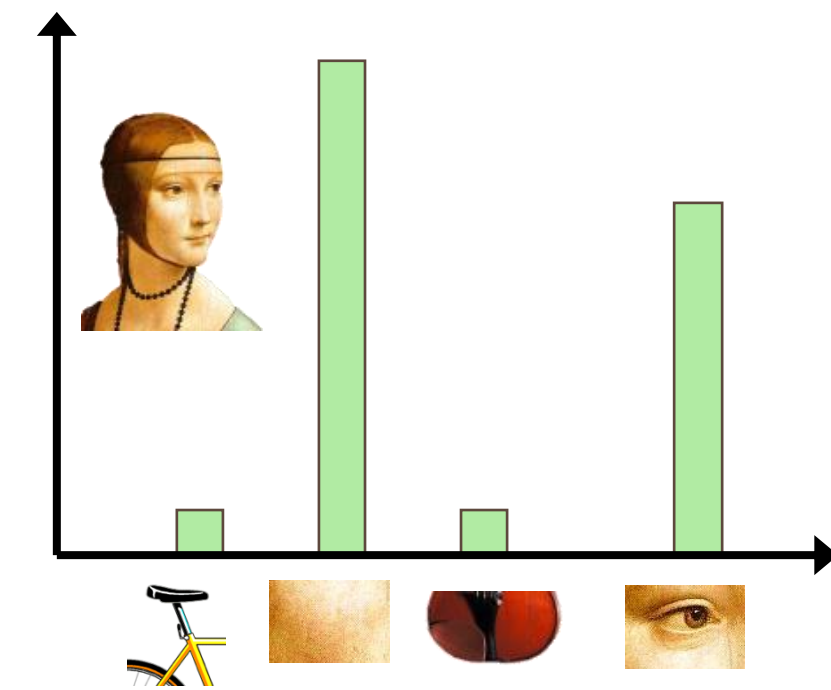
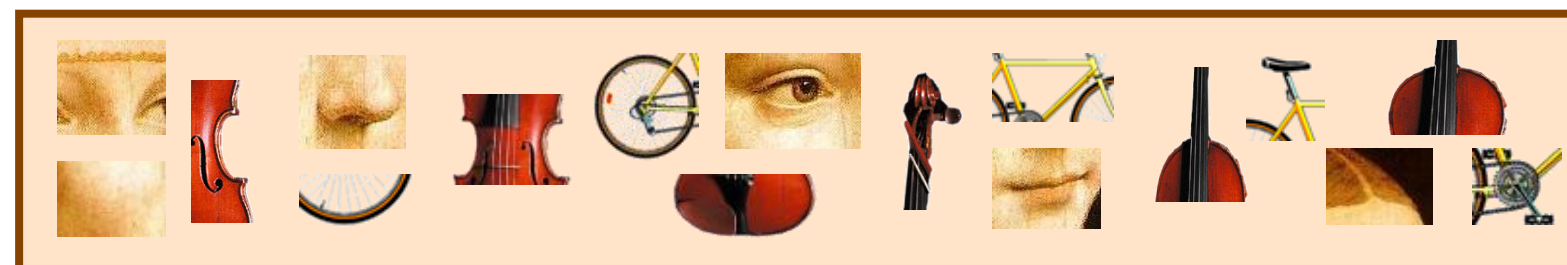


Bag-of-features models



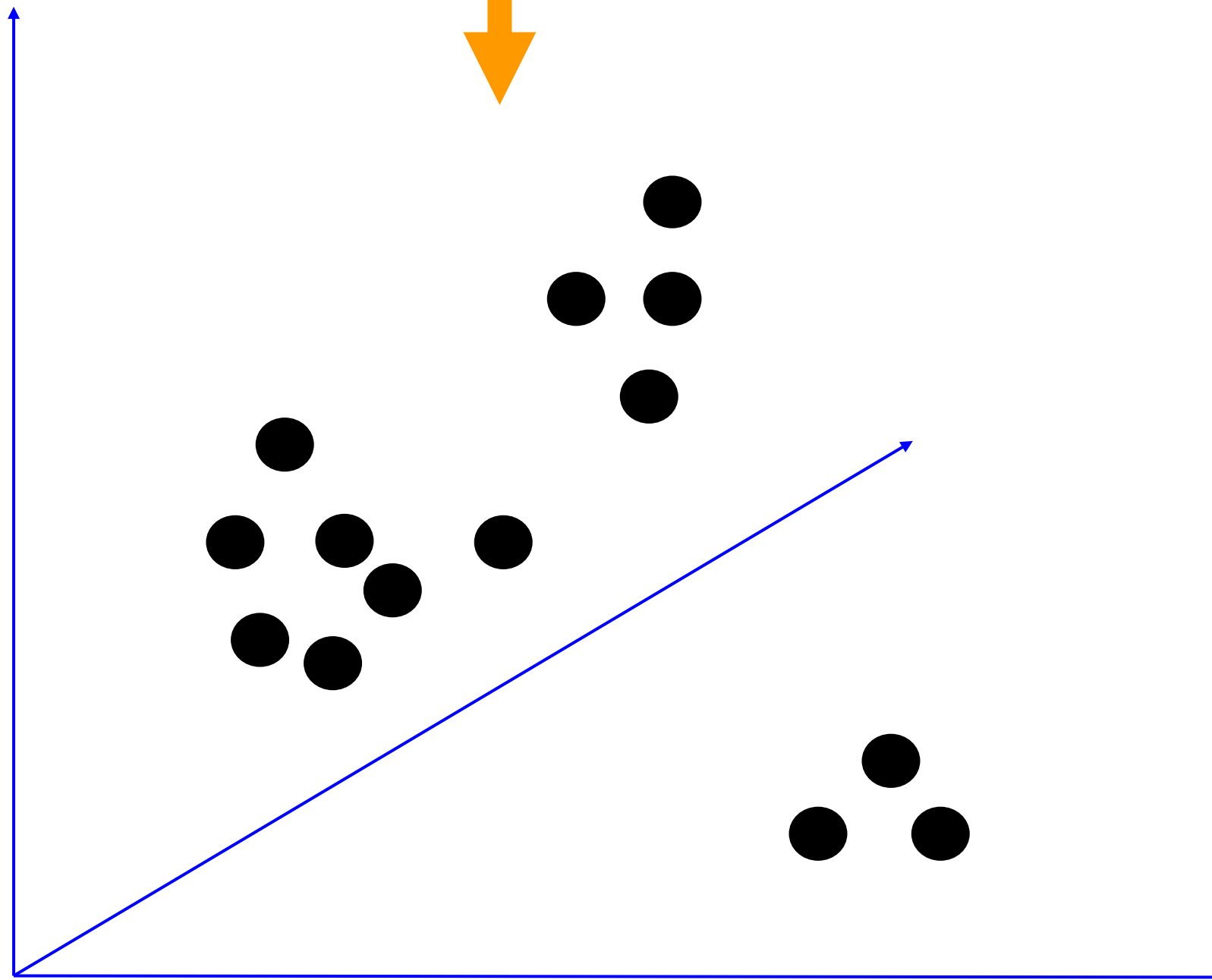
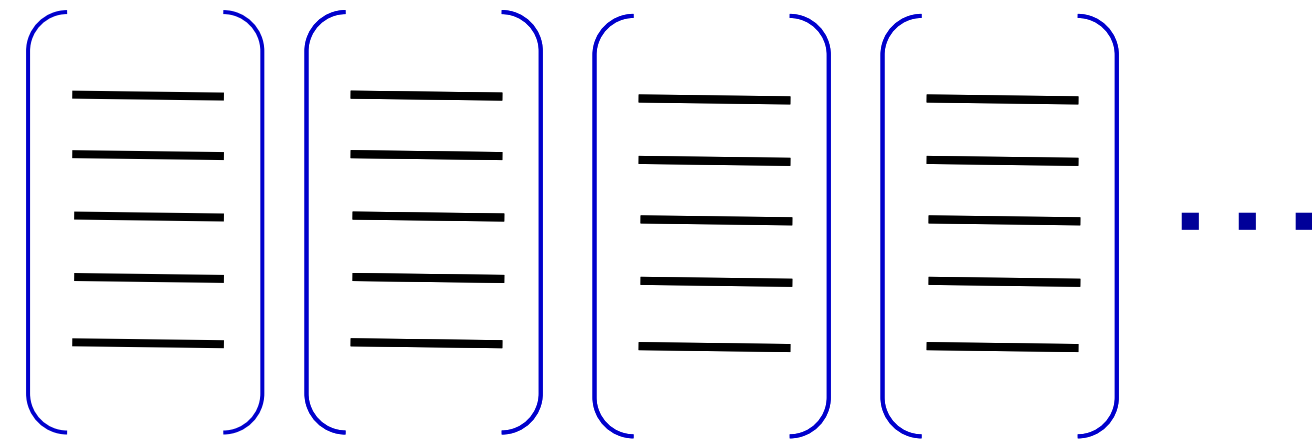
Bags of visual words

- Summarize entire image based on its distribution (histogram) of word occurrences.
- Analogous to bag of words representation commonly used for documents.

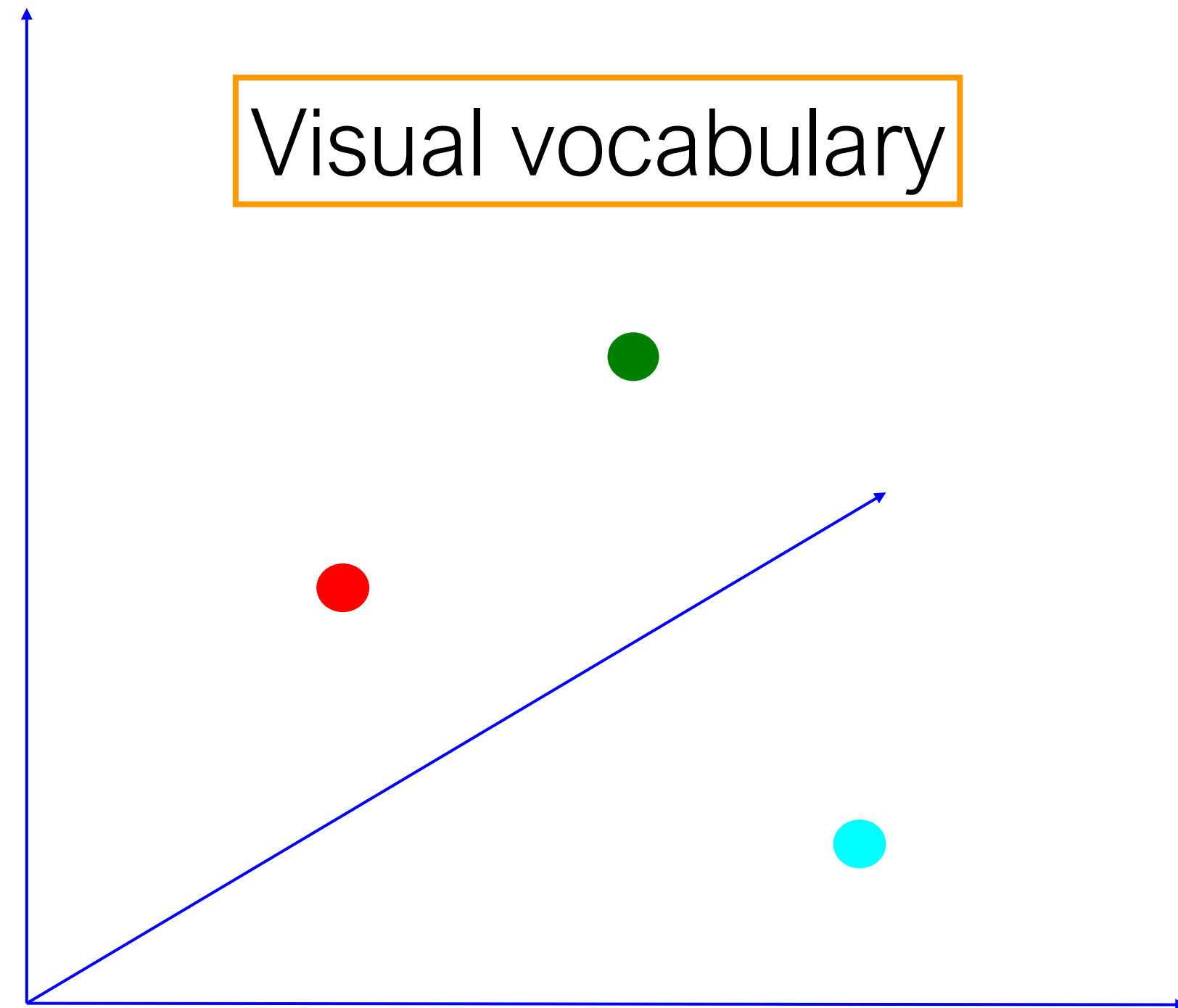


Visual vocabulary

Local
descriptors



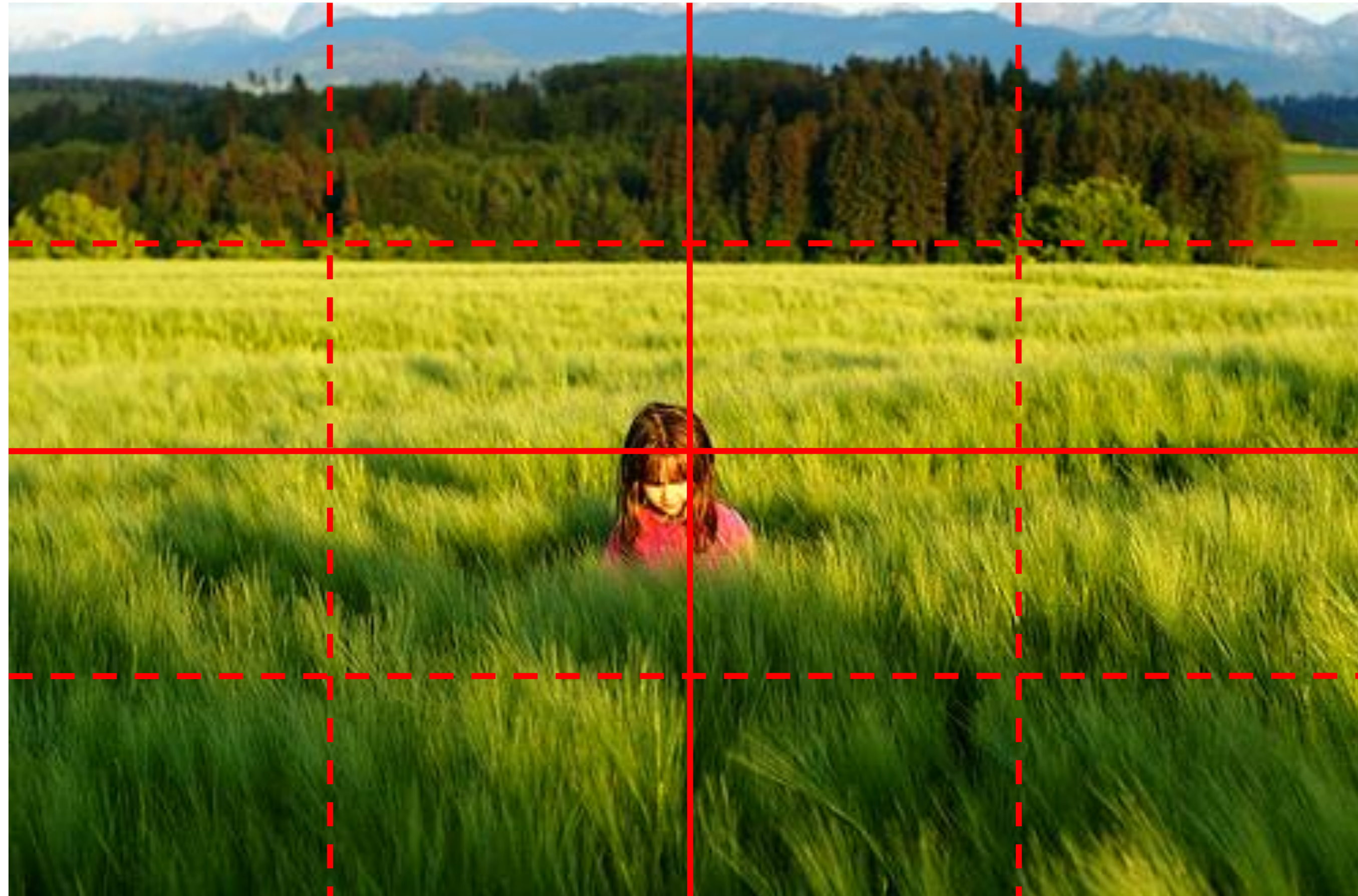
Visual vocabulary



Clustering

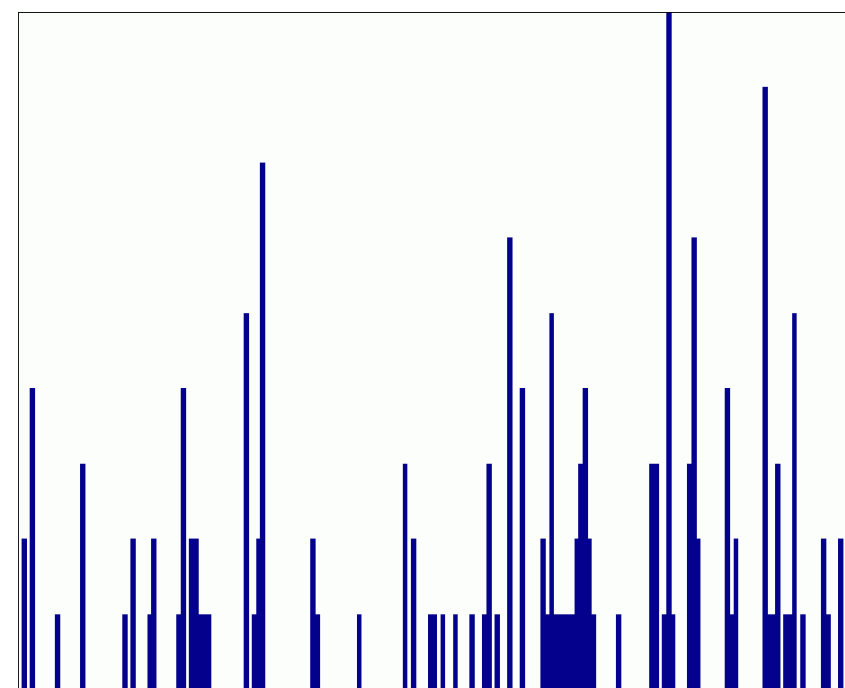


Spatial pyramid representation



Compute histogram in each spatial bin

Spatial pyramid representation

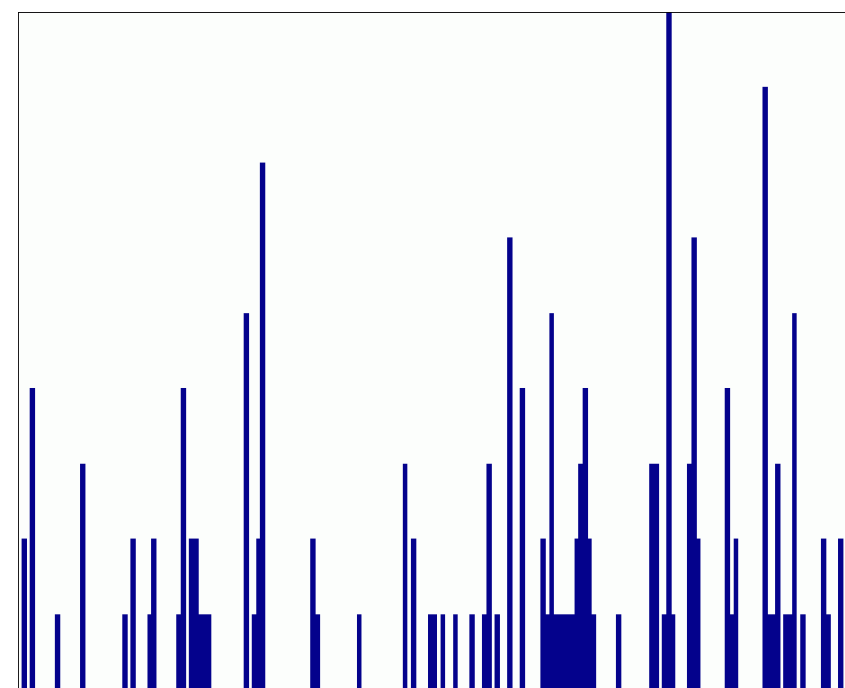
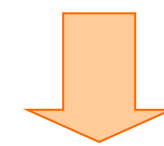
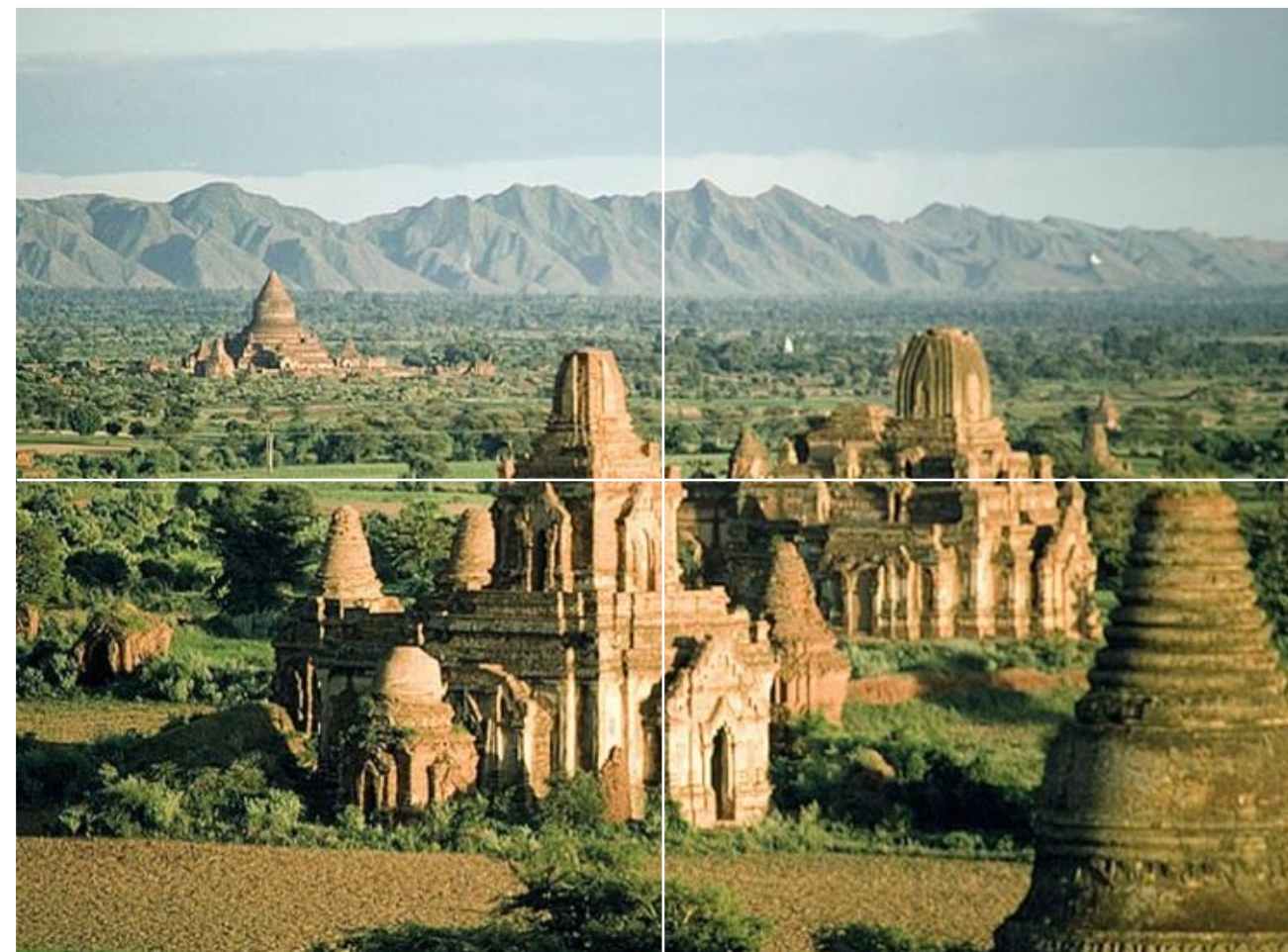


level 0

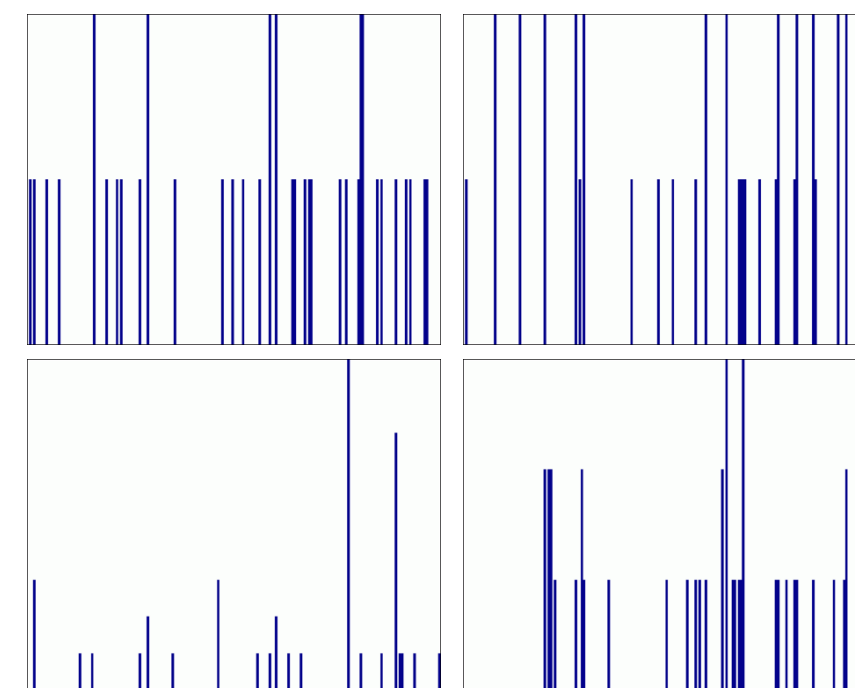
Extension of a bag of words

Locally orderless representation at
several levels of resolution

Spatial pyramid representation

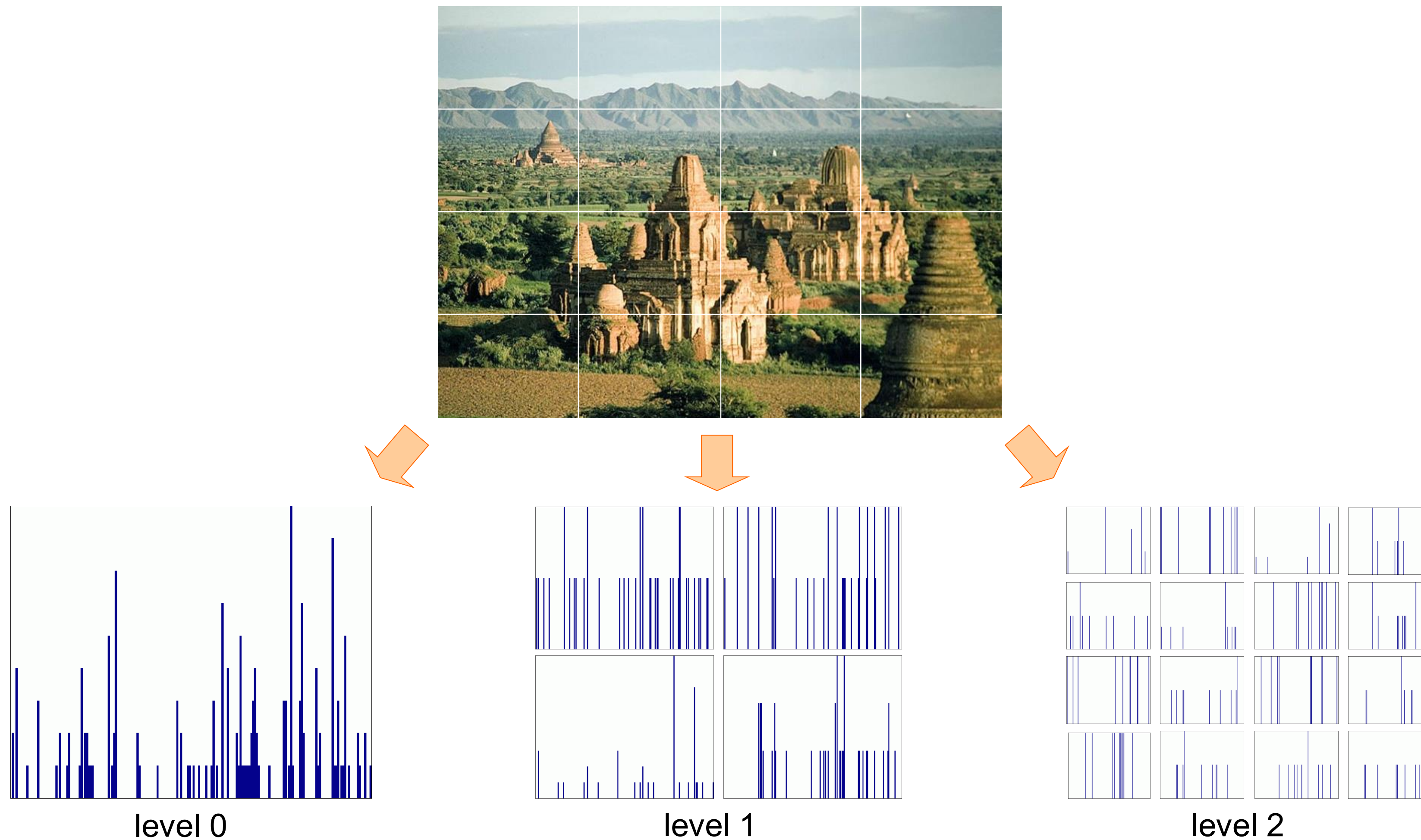


level 0

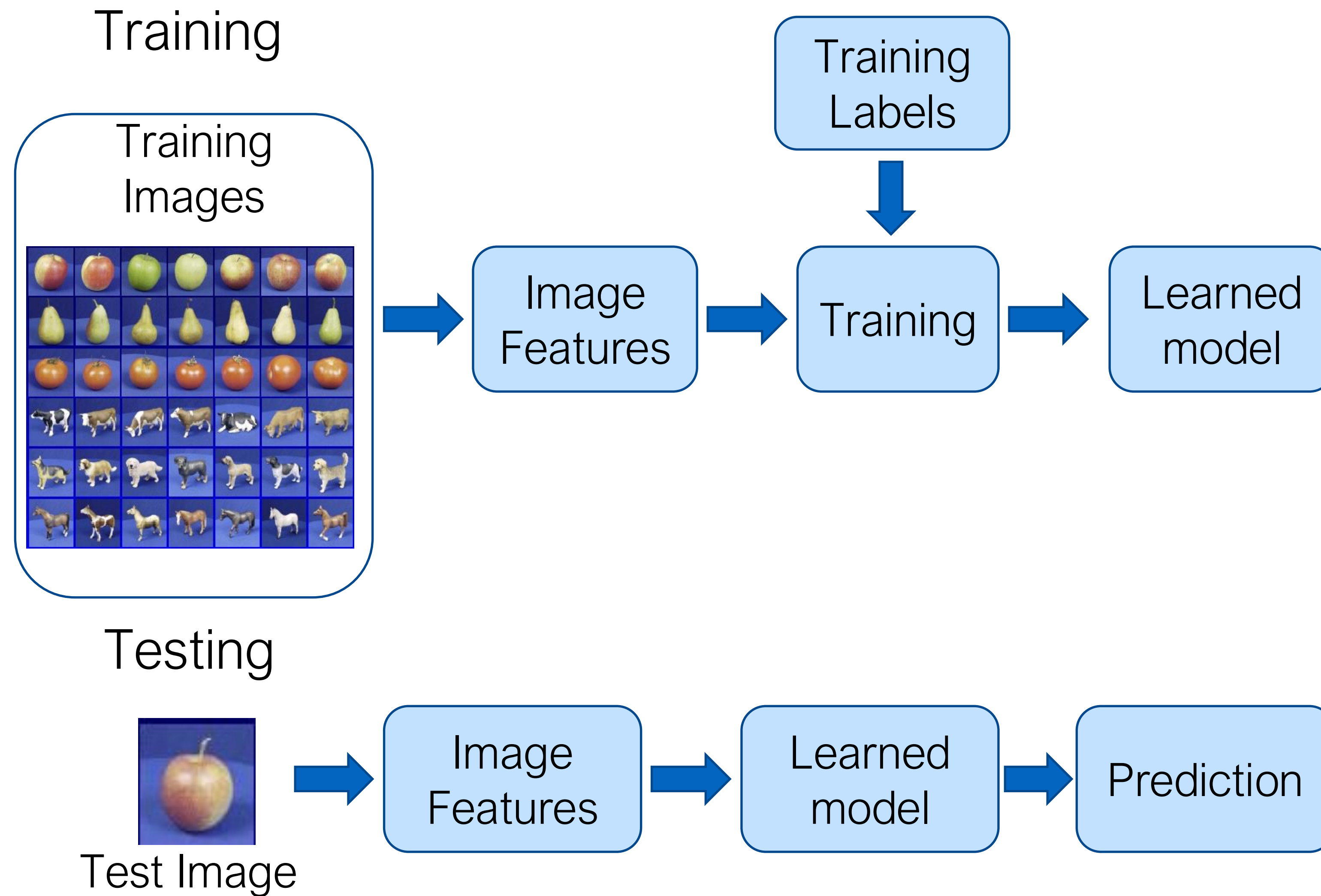


level 1

Spatial pyramid representation



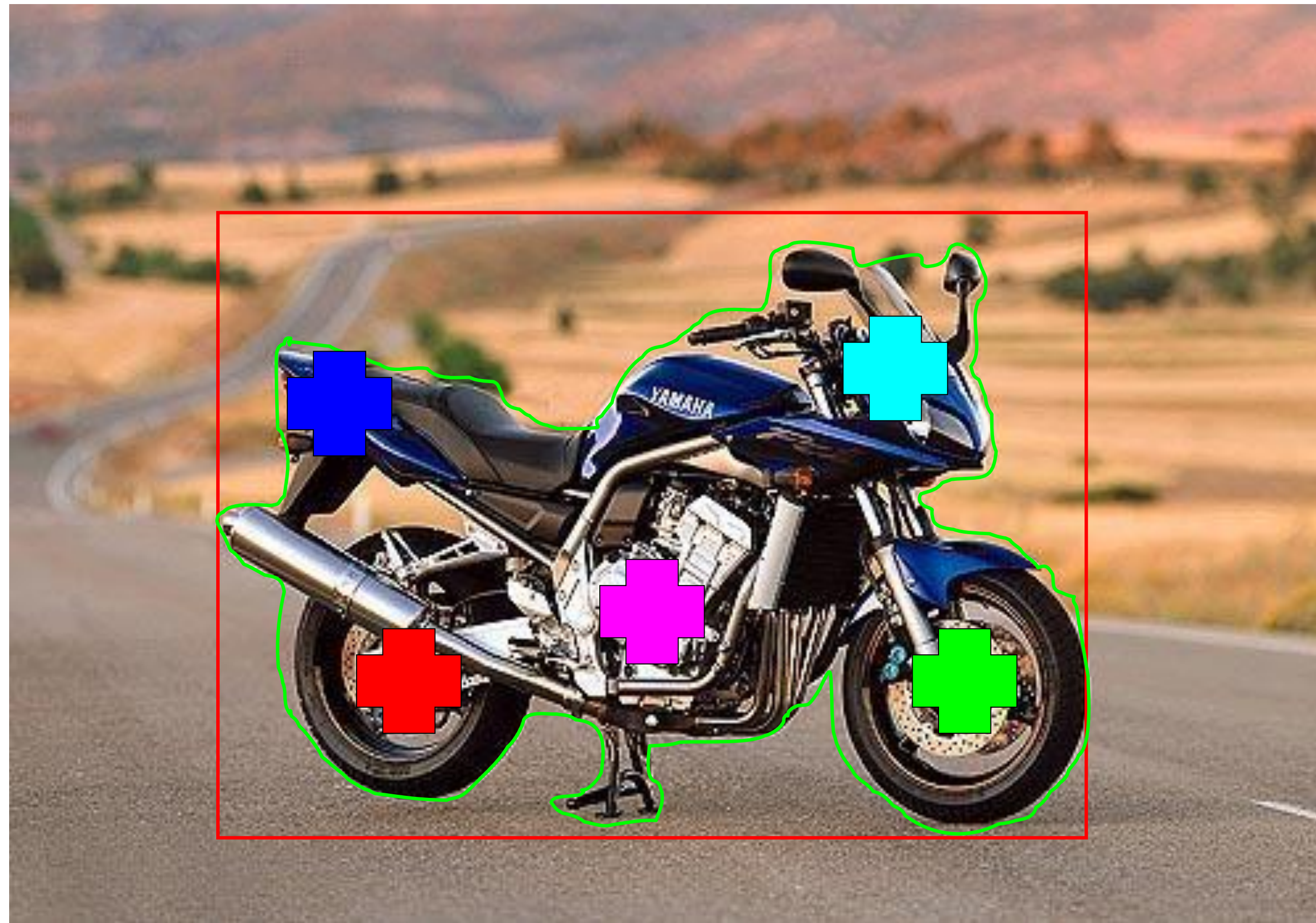
Recognition in computer vision



Training labels

Images in the training set must be annotated with the “correct answer” that the model is expected to produce

Contains a motorbike



Many ML classifiers to choose from...

K-nearest neighbor

SVM

Naïve Bayes

Bayesian network

Logistic regression

Randomized Forests

Boosted Decision Trees

Restricted Boltzmann Machines

Neural networks

Convolutional Neural Networks

...

Which one is the
best?

Dataset split

Training
Images



- Train classifier

Validation
Images



- Measure error
- Tune model hyperparameters

Testing
Images



- Secret labels
- Measure error

Random training/validation splits = cross validation

k-fold

